

CARTAGENA, Colombia

WMO: 800220

Lat: 10.442N

Lon: 75.513W

Elev: 1

StdP: 101.31

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	23.9	20.8	15.4	25.8	21.1	15.7	25.9	9.6	28.7	8.9	28.8	1.4	20	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.6	33.2	27.6	32.9	27.5	32.2	27.1	28.5	31.8	28.1	31.4	27.9	31.0	4.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	30.5	27.2	22.9	30.4	27.0	22.7	30.2	92.1	32.4	90.7	31.8	89.2	31.3	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.3	6.4	21.0	35.2	2.0	1.5	19.6	36.3	18.4	37.1	17.3	37.9	15.8	39.0		
			20.0	29.6	2.0	0.6	18.6	30.1	17.4	30.5	16.3	30.8	14.8	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	27.4	27.6	27.9	28.5	28.8	29.0	28.9	28.9	28.9	28.5	28.3	28.1
	DBStd	1.08	0.76	0.88	0.84	0.87	0.96	1.00	0.92	1.01	1.09	1.07	1.00	0.86
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6718	541	492	554	556	584	569	586	567	573	549	561	
	CDD18.3	3676	282	258	296	306	326	319	327	328	317	299	303	
	CDH23.3	41378	2790	2561	3041	3449	3864	3853	3879	3911	3796	3629	3364	3241
	CDH26.7	14646	811	751	894	1172	1462	1534	1498	1528	1498	1307	1138	1052
Wind	WSAvg	2.7	3.2	3.5	3.8	3.2	2.4	2.3	2.6	2.4	2.2	2.2	2.2	2.7
Precipitation	PrecAvg	1103	5	1	3	25	99	103	108	134	150	248	147	43
	PrecMax	2438	108	9	50	110	293	283	379	428	369	605	579	208
	PrecMin	401	0	0	0	0	5	9	4	22	42	51	35	0
	PrecStd	442	18	2	9	30	70	66	87	87	74	129	105	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.2	32.2	33.1	33.2	33.9	33.8	33.4	33.9	33.2	33.1	33.0
		MCWB	26.0	25.8	26.3	27.3	27.8	28.1	27.5	27.7	28.1	27.8	27.6	26.9
	2%	DB	31.2	31.2	31.4	32.1	32.2	32.9	32.8	32.8	32.9	32.2	32.1	32.0
		MCWB	26.0	25.8	26.2	26.9	27.4	27.7	27.4	27.6	27.8	27.3	27.1	26.6
	5%	DB	30.8	30.9	30.9	31.2	31.9	32.1	32.1	32.0	32.1	31.8	31.2	31.2
		MCWB	25.9	25.7	26.1	26.7	27.2	27.3	27.1	27.2	27.3	27.1	26.9	26.4
	10%	DB	30.1	30.1	30.1	30.9	31.1	31.6	31.5	31.2	31.2	31.0	30.9	30.8
		MCWB	25.6	25.4	25.8	26.6	27.0	27.1	27.0	27.0	27.1	26.9	26.7	26.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.1	27.6	28.2	28.7	28.7	28.5	28.6	28.7	28.7	28.2	28.0
		MCDB	30.7	30.7	30.5	31.4	31.8	32.2	31.8	31.9	32.2	31.8	31.4	30.8
	2%	WB	26.5	26.6	26.9	27.5	28.1	28.2	28.0	28.1	28.2	28.0	27.7	27.3
		MCDB	30.2	30.1	30.2	30.7	31.2	31.6	31.5	31.3	31.4	31.0	30.6	30.4
	5%	WB	26.2	26.1	26.5	27.1	27.6	27.7	27.6	27.7	27.8	27.5	27.2	26.9
		MCDB	29.7	29.7	29.7	30.3	30.6	31.0	31.0	30.8	30.8	30.4	30.0	30.1
	10%	WB	25.8	25.7	26.1	26.7	27.2	27.4	27.2	27.3	27.5	27.1	26.9	26.6
		MCDB	29.2	29.2	29.3	29.7	30.1	30.5	30.4	30.3	30.3	29.8	29.6	29.5
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.2	5.6	5.2	5.2	5.6	5.9	5.7	5.6	5.5	5.4	6.0
		MCDBR	6.8	6.8	6.1	5.7	5.8	6.2	6.4	6.1	6.2	6.1	5.9	6.6
		MCWBR	3.1	3.1	2.9	2.7	2.7	2.9	3.0	2.9	2.9	2.8	2.7	2.9
	5% WB	MCDBR	6.3	6.2	5.6	5.4	5.5	5.9	6.1	5.9	5.9	5.9	5.5	6.1
		MCWBR	3.1	3.1	2.9	2.7	2.8	3.0	3.0	3.0	2.9	3.0	2.7	2.9
Clear-Sky Solar Irradiance	taub	0.401	0.434	0.470	0.524	0.485	0.459	0.455	0.455	0.448	0.411	0.402	0.403	
	taud	2.496	2.393	2.292	2.124	2.275	2.356	2.376	2.378	2.369	2.507	2.534	2.516	
	Ebn at Noon	906	886	857	802	818	829	836	848	863	896	900	896	
	Edh at Noon	108	123	138	162	136	124	122	124	126	108	103	104	
All-Sky Solar Radiation	RadAvg	5.61	5.97	5.89	5.66	5.13	5.33	5.53	5.45	5.13	4.68	4.60	5.04	
	RadStd	0.14	0.27	0.32	0.32	0.37	0.34	0.34	0.34	0.25	0.26	0.42	0.39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.35	N/A	N/A	+0.71	+0.41	+0.20	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	+0.32	N/A	N/A	+0.62	+0.32	+0.14	N/A	N/A	+122	+122

Nomenclature: See separate page